

Composing the parallel dimensions (concrete)

one PP stage

a DP $\times$ TP $\times$ EP mesh of GPUs (pipeline divides the model into stages)

stage boundary $\rightarrow$ next stage (P2P activations)

W (weight matrix) — TP-sharded across the 2 TP rows

GPU 1

(DP rank 1 $\cdot$ TP rank 1)

GPU 2

(DP rank 2 $\cdot$ TP rank 1)

GPU 3

(DP rank 1 $\cdot$ TP rank 2)

GPU 4

(DP rank 2 $\cdot$ TP rank 2)

data batch — DP-sharded across the 2 DP columns

EP: MoE experts sharded across all 4 GPUs — token $\rightarrow$ expert = all-to-all

experts 1-15 $\cdot$ each GPU holds its slice; attention/shared weights replicated

dimension

PARTITIONED

REPLICATED

COMMUNICATION



TP
PP
DP
EP
CP (opt)

W rows
layers
data batch
MoE experts
token sequence

activations
boundary activations
model + optimizer
attn + weights
model weights

all-reduce
P2P
all-reduce
all-to-all
ring